Integration by Parts: Complete Tutorial with 20 Practice Problems

Calculus Tutorial

Contents

1 Introduction to Integration by Parts

1.1 What is Integration by Parts?

Integration by parts is used when an integral is a **product of two functions**, for example:

- $x \cdot e^x$
- $x \cdot \sin x$
- $\ln x$
- $x \cdot \ln x$

1.2 The Basic Idea

In plain words:

> You "split" the integral into two parts, make one part simpler through differentiation, and then integrate what remains.

2 The Integration by Parts Formula

The formula is derived from the product rule for differentiation:

$$\int u\,dv = uv - \int v\,du$$

2.1 How to Use the Formula

1. Choose **one part** as u (something that becomes simpler when differentiated)
2. The rest becomes dv
3. Differentiate u to get du
4. Integrate dv to get v
5. Substitute into the formula

3 How to Choose u (The LIATE Rule)

Letter	Type of Function	Example
L	Logarithmic	$\ln x$, $\log x$
I	Inverse Trigonometric	$\tan^{-1} x$, $\sin^{-1} x$
A	Algebraic (Polynomials)	x, x^2, x^3
T	Trigonometric	$\sin x$, $\cos x$
E	Exponential	e^x, 2^x

Rule: Choose u from the **earliest** category in LIATE that appears in the integrand.

4 Step-by-Step Examples

4.1 Example 1: Basic Case

$$\int xe^x\,dx$$

Solution:

$$\begin{aligned} \text{Let } u = x \quad &\Rightarrow \quad du = dx \\ dv = e^x dx \quad &\Rightarrow \quad v = e^x \end{aligned}$$

$$\begin{aligned} \int xe^x\,dx &= xe^x - \int e^x dx \\ &= xe^x - e^x + C \\ &= e^x(x-1) + C \end{aligned}$$

4.2 Example 2: Logarithmic Function

$$\int \ln x\,dx$$

Solution:

$$\begin{aligned}
\text{Let } u = \ln x \quad &\Rightarrow \quad du = \frac{1}{x}dx \\
dv = dx \quad &\Rightarrow \quad v = x \\
\int \ln x\,dx &= x\ln x - \int x \cdot \frac{1}{x}dx \\
&= x\ln x - \int 1dx \\
&= x\ln x - x + C
\end{aligned}$$

5 20 Practice Problems with Solutions

5.1 Easy Level

1. $\int x\,dx$

   Solution: $\frac{x^2}{2} + C$

2. $\int xe^x\,dx$

   Solution: $e^x(x-1) + C$

3. $\int x\sin x\,dx$

   Solution:

$$\begin{aligned}
u = x, \quad & dv = \sin x\,dx \\
du = dx, \quad & v = -\cos x \\
\int x\sin x\,dx &= -x\cos x + \int \cos x\,dx \\
&= -x\cos x + \sin x + C
\end{aligned}$$

4. $\int x\cos x\,dx$

   Solution:

$$\begin{aligned}
u = x, \quad & dv = \cos x\,dx \\
du = dx, \quad & v = \sin x \\
\int x\cos x\,dx &= x\sin x - \int \sin x\,dx \\
&= x\sin x + \cos x + C
\end{aligned}$$

5. $\int \ln x\,dx$

   Solution: $x\ln x - x + C$

5.2 Medium Level

6. $\int x^2e^x\,dx$

Solution:

$$u = x^2, \quad dv = e^x\,dx$$
$$du = 2x\,dx, \quad v = e^x$$
$$\begin{aligned}\int x^2 e^x\,dx &= x^2 e^x - 2\int x e^x\,dx \\ &= x^2 e^x - 2[e^x(x-1)] + C \\ &= e^x(x^2 - 2x + 2) + C\end{aligned}$$

7. $\int x^2 \sin x\,dx$

Solution:

$$u = x^2, \quad dv = \sin x\,dx$$
$$du = 2x\,dx, \quad v = -\cos x$$
$$\begin{aligned}\int x^2 \sin x\,dx &= -x^2\cos x + 2\int x\cos x\,dx \\ &= -x^2\cos x + 2[x\sin x + \cos x] + C \\ &= -x^2\cos x + 2x\sin x + 2\cos x + C\end{aligned}$$

8. $\int x^3 e^x\,dx$

Solution:

$$u = x^3, \quad dv = e^x\,dx$$
$$du = 3x^2\,dx, \quad v = e^x$$
$$\begin{aligned}\int x^3 e^x\,dx &= x^3 e^x - 3\int x^2 e^x\,dx \\ &= x^3 e^x - 3e^x(x^2 - 2x + 2) + C \\ &= e^x(x^3 - 3x^2 + 6x - 6) + C\end{aligned}$$

9. $\int x \ln x\,dx$

Solution:

$$u = \ln x, \quad dv = x\,dx$$
$$du = \frac{1}{x}dx, \quad v = \frac{x^2}{2}$$
$$\begin{aligned}\int x\ln x\,dx &= \frac{x^2}{2}\ln x - \int \frac{x^2}{2}\cdot\frac{1}{x}dx \\ &= \frac{x^2}{2}\ln x - \frac{1}{2}\int x\,dx \\ &= \frac{x^2}{2}\ln x - \frac{x^2}{4} + C\end{aligned}$$

10. $\int \ln(x^2)\,dx$

Solution:

$$\begin{aligned}\int \ln(x^2)\,dx &= \int 2\ln x\,dx \\ &= 2(x\ln x - x) + C \\ &= 2x\ln x - 2x + C\end{aligned}$$

5.3 Trigonometric and Exponential

11. $\int e^x \sin x \, dx$

Solution: (Use integration by parts twice)

$$\begin{aligned}
\text{Let } I &= \int e^x \sin x \, dx \\
u &= e^x, \quad dv = \sin x \, dx \\
du &= e^x \, dx, \quad v = -\cos x \\
I &= -e^x \cos x + \int e^x \cos x \, dx
\end{aligned}$$

Apply integration by parts again to $\int e^x \cos x \, dx$:

$$\begin{aligned}
u &= e^x, \quad dv = \cos x \, dx \\
du &= e^x \, dx, \quad v = \sin x \\
\int e^x \cos x \, dx &= e^x \sin x - \int e^x \sin x \, dx \\
&= e^x \sin x - I
\end{aligned}$$

Therefore:

$$\begin{aligned}
I &= -e^x \cos x + e^x \sin x - I \\
2I &= e^x(\sin x - \cos x) \\
I &= \frac{e^x}{2}(\sin x - \cos x) + C
\end{aligned}$$

12. $\int e^x \cos x \, dx$

Solution:

$$\begin{aligned}
\text{Let } I &= \int e^x \cos x \, dx \\
\text{Using similar method: } I &= \frac{e^x}{2}(\sin x + \cos x) + C
\end{aligned}$$

13. $\int xe^{2x} \, dx$

Solution:

$$\begin{aligned}
u &= x, \quad dv = e^{2x} \, dx \\
du &= dx, \quad v = \frac{1}{2}e^{2x} \\
\int xe^{2x} \, dx &= \frac{x}{2}e^{2x} - \frac{1}{2}\int e^{2x} \, dx \\
&= \frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C \\
&= \frac{e^{2x}}{4}(2x - 1) + C
\end{aligned}$$

14. $\int x \sin 2x \, dx$

Solution:

$$\begin{aligned} u = x, &\quad dv = \sin 2x\,dx \\ du = dx, &\quad v = -\frac{1}{2}\cos 2x \end{aligned}$$

$$\begin{aligned} \int x\sin 2x\,dx &= -\frac{x}{2}\cos 2x + \frac{1}{2}\int \cos 2x\,dx \\ &= -\frac{x}{2}\cos 2x + \frac{1}{4}\sin 2x + C \end{aligned}$$

15. $\int x\cos 3x\,dx$

Solution:

$$\begin{aligned} u = x, &\quad dv = \cos 3x\,dx \\ du = dx, &\quad v = \frac{1}{3}\sin 3x \end{aligned}$$

$$\begin{aligned} \int x\cos 3x\,dx &= \frac{x}{3}\sin 3x - \frac{1}{3}\int \sin 3x\,dx \\ &= \frac{x}{3}\sin 3x + \frac{1}{9}\cos 3x + C \end{aligned}$$

5.4 Challenging Problems

16. $\int x^2\ln x\,dx$

Solution:

$$\begin{aligned} u = \ln x, &\quad dv = x^2\,dx \\ du = \frac{1}{x}dx, &\quad v = \frac{x^3}{3} \end{aligned}$$

$$\begin{aligned} \int x^2\ln x\,dx &= \frac{x^3}{3}\ln x - \int \frac{x^3}{3}\cdot\frac{1}{x}dx \\ &= \frac{x^3}{3}\ln x - \frac{1}{3}\int x^2\,dx \\ &= \frac{x^3}{3}\ln x - \frac{x^3}{9} + C \end{aligned}$$

17. $\int \ln^2 x\,dx$

Solution:

$$\begin{aligned} u = \ln^2 x, &\quad dv = dx \\ du = \frac{2\ln x}{x}dx, &\quad v = x \end{aligned}$$

$$\begin{aligned} \int \ln^2 x\,dx &= x\ln^2 x - 2\int \ln x\,dx \\ &= x\ln^2 x - 2(x\ln x - x) + C \\ &= x\ln^2 x - 2x\ln x + 2x + C \end{aligned}$$

18. $\int \tan^{-1} x\,dx$

Solution:

$$\begin{aligned}
u &= \tan^{-1} x, \quad dv = dx \\
du &= \frac{1}{1+x^2} dx, \quad v = x \\
\int \tan^{-1} x \, dx &= x \tan^{-1} x - \int \frac{x}{1+x^2} dx \\
&= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C
\end{aligned}$$

19. $\int xe^{-x} \, dx$

Solution:

$$\begin{aligned}
u &= x, \quad dv = e^{-x} \, dx \\
du &= dx, \quad v = -e^{-x} \\
\int xe^{-x} \, dx &= -xe^{-x} + \int e^{-x} \, dx \\
&= -xe^{-x} - e^{-x} + C \\
&= -e^{-x}(x+1) + C
\end{aligned}$$

20. $\int x^2 e^{-x} \, dx$

Solution:

$$\begin{aligned}
u &= x^2, \quad dv = e^{-x} \, dx \\
du &= 2x \, dx, \quad v = -e^{-x} \\
\int x^2 e^{-x} \, dx &= -x^2 e^{-x} + 2 \int xe^{-x} \, dx \\
&= -x^2 e^{-x} + 2[-e^{-x}(x+1)] + C \\
&= -e^{-x}(x^2 + 2x + 2) + C
\end{aligned}$$

6 Memory Trick

$\boxed{\textbf{Product inside integral? } \rightarrow \textbf{ Think Integration by Parts!}}$

7 Summary

Integration by parts is a powerful technique for integrating products of functions. Remember:

- Use the formula: $\int u \, dv = uv - \int v \, du$
- Choose u using the LIATE rule
- Sometimes you need to apply the method multiple times
- Practice is key to recognizing when and how to use this method